Journal Paper Summary (25 points total)

Paper 1 Title	Impact of data visualization on decision-making and its implications for public health practice: a systematic literature review
Author Name(s)	Seungeun Park, Petty Bekemeier, Abraham Flaxman, Melinda Schultz
Paper 2 Title	What is visualization?
Author Name(s)	Lev Manovich
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1. According to both papers, what is Information Visualization? Discuss the two principles that information visualization rests on. [5 points]

Focus of this question
• This question encourages you to evaluate the arguments, evidence, assumptions, and conclusions about key issues (i.e. think critically about the paper) • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts) • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc) during class or in assignments

Write your answer below:

According to the first paper “Impact of data visualization on decision-making and its implication for public health practice” the authors define data visualization as “visual representations such as graphs and maps used to understand, communicate, and support collaborative engagement and decision-making with data and information. Data visualizations can help people understand data, gain insight, answer specific questions, and discover underlying facts.” They do not give nearly as much account as Manovich does in his paper about defining visualization, but the conclusion of this paper found that data visualization “suggests positive effects of data visualization depending on the control of confounding factors.”

According to the paper “What is visualization?” Manovich expresses difficulty in creating one overarching definition about visualization. Manovich states that information visualization relied on two key principles: reduction and spatial variables.

Manovich describes information visualization using reduction as “us[ing] graphical primitives such as points, straight lines, curves and simple geometric shapes to stand in for objects and relations between them.” Manovich argues that by using graphical primitives, information visualization can uncover patterns and structures that these primitives represent. These patterns and structures help us map the properties of data that we are most interested in, letting other less important dimensions be represented through other visual mediums such as tone, shading, etc.

Manovich believes that information visualization mostly lies in the principle of spatial variables which is described as “reserving spatial arrangement (layout) for the most important dimensions of the data and using other visual variables for remaining dimensions.” Manovich argues that this is the most important aspect of visualization as he points out that sixteenth century paint techniques of spatially arranging parts as well as Asian “ink and wash painting” hardly rely on colors, textures, and other visual parameters, and highlights that these techniques “explore the contrast between objects’ contours, their spatial arrangements and different types of brushstrokes.” Manovich points out that “The creators of visualizations follow human visual perception that also privileges spatial arrangements of parts of a scene over its other visual properties.” which may be why this is one of the most important principles of visualization.

2. In what other ways can information visualization support public health practice? How have the fields of computer science and information science affected the development of information visualization? [5 points]

Focus of this question
• This question encourages you to connect your learning to "real world" issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

According to the research paper, information visualization can support public health by "answering questions that previously were unanswerable". In addition, the paper also mentions that public health leaders in various degrees of power can use information visualization for "resource allocation and public health planning" where crucial decisions need the best data and interpretation of that data.

The fields of computer science and information science have "developed data visualization tools and described the utility and usability of these tools in their field." The expanded usability of these tools has made it so that data visualization can be accessible even to fields outside of the scope of computer science and information science, which can be used to support more intricate data analysis and better ways to present data to key stakeholders, public health leaders, etc.

3. One of the papers discusses the advantages that visualization contributes to a "wide variety of fields" – according to the studies the authors reviewed. List and discuss two (2) of these advantages. What was the key limitation across most of the studies the authors reviewed? [15 points]

Focus of this question
• This question encourages you to connect your learning to "real world" issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to reflect on what you are learning • This question encourages you to contribute your ideas and thoughts • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc) during class or in assignments • This question challenges you to develop and present your own knowledge, comprehension, and conceptual understanding

Write your answer below:

According to this paper, visualization brings advantages such as "increasing the amount of information delivered" and "decreasing the cognitive and intellectual burden to interpret information for decision-making". The studies that the authors reviewed were "limited theoretically and methodologically". The paper states that most studies that do not present any theories used for their investigations, and the diversity of the academic disciplines and topics made it difficult to find consistency in methods and outcome measures. There were no guidelines on behavioral studies, and some studies "did not appear to report the essential elements of an experimental study". Overall, every study had a different way of defining visualization and measuring it which makes it hard to consolidate and do a mass review on when no one can agree on one central definition of information visualization or way to go about measuring it.